Question: 4 (CLO # 6) **5 Marks**

a) Draw $i_D - v_{DS}$ characteristics o[illegible] annel enhancemen[illegible] Show at least 4 curves for 4 different values of V_{GS})

   i. Identify the operating regions of NMOS on the charac[illegible] curve.

   ii. Give the Operating conditions for V_{DS} and V_{GS} and expression for current in each region.

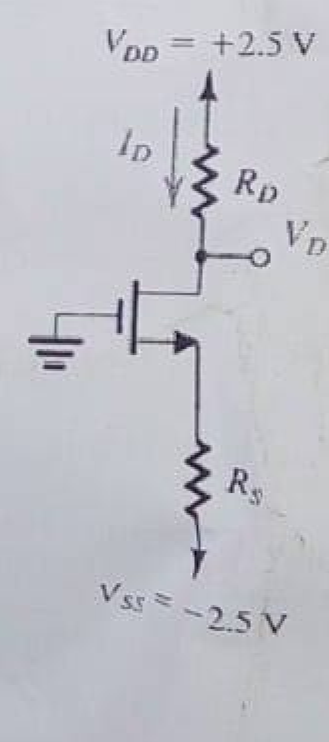
   iii. Mark the region on characteristic curve where MOS can be used as a resistor.

b) For an NMOS transistor, for which $V_t = 0.8V$ operating with $V_{GS} = 4V$, what is the value of V_{DS} for which channel remains uniform?

c) An NMOS transistor, fabricated with $W = 100\mu m$ and $L = 5\mu m$ in a technology for which $k_n' = 50\mu A/V^2$ and $V_t = 1V$, is to be operated at very low values of V_{DS} as a linear amplifier. For V_{GS} varying from 1.1V to 11V, what range of resistor values can be obtained?

Question: 5 (CLO # 7) **10 Marks**

Design the circuit in figure below for the following requirements: $V_{DD} = -V_{SS} = 2.5V$, $I_D = 0.3mA$, $V_D = +0.4V$. Let $V_t = 1V$, $\mu_n C_{ox} = 60\ \mu A/V^2$, $W/L = 120\ \mu m/3\ \mu m$.

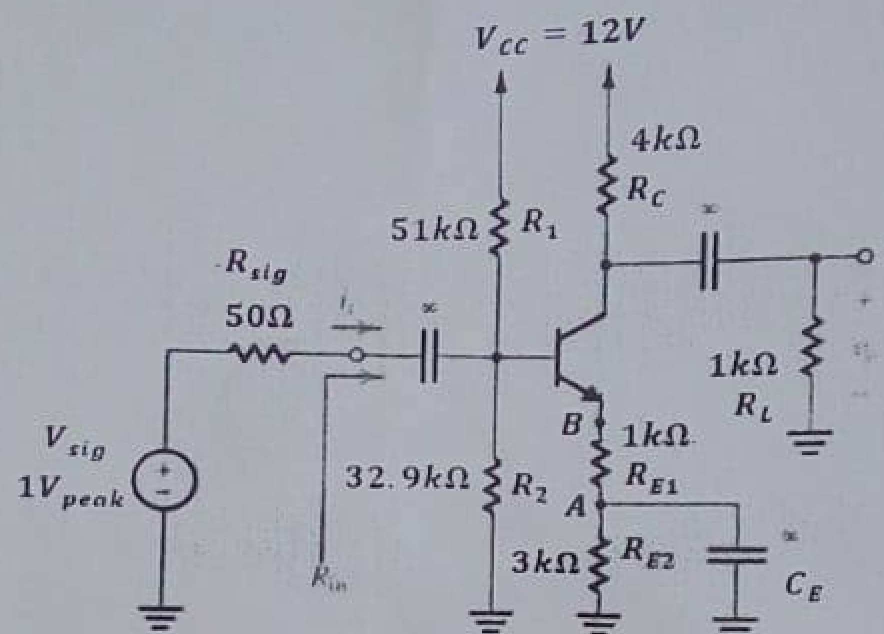


Question: 6 (PLO # 2, CLO#5) 25 Marks

Consider the following amplifier circuit with split emitter resistors

$R_1 = 51k\Omega$, $R_2 = 32.9k\Omega$, $R_C = 4k\Omega$, $R_{E_1} = 1k\Omega$, $R_{E_2} = 3k\Omega$, $V_{CC} = 12V$, $R_{sig} = 50\Omega$

$V_{sig} = 1V_{peak}$, $R_L = 1k\Omega$, $\beta = 100$



a) Find V_C (dc) and I_C (dc) of the above circuit. (S2.1)

b) Draw small signal equivalent model of the circuit (S2.1)

i. When emitter bypass capacitor C_E is connected to point A as shown in the figure above.

ii. When emitter bypass capacitor C_E is connected to point B (i.e. directly with emitter)

iii. No emitter bypass capacitor in the circuit.

c) Write the mathematical expressions for overall voltage gain G_V, input resistance and output resistance for all the three cases in part b? (S2.2)

d) Calculate the overall voltage gain G_V, input resistance and output resistance for all the three cases in part b? (S2.3)

e) Compare all three cases in terms of voltage gain, input and output resistances? Explain your results from the calculations of part d? Conclude which one is a better voltage amplifier and why? (S2.4)

National University of Computer and Emerging Sciences, Lahore Campus

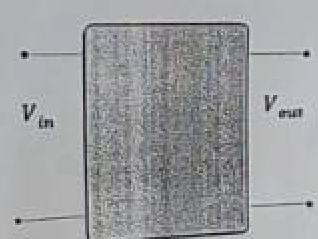
Course Name:	Electronic Devices & Circuits	Course Code:	EE225
Program:	Electrical Engineering	Semester:	Fall 2018
Duration:	180 Minutes	Total Marks:	100
Paper Date:	27-12-2018	Weight	50%
Section:	ALL	Page(s):	4
Exam Type:	Final		

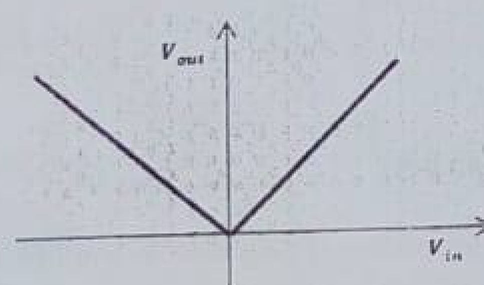
Student : Name: [redacted] Roll No. [redacted] Section:

Instruction/Notes: Carefully read the paper and attempt each question (only once). Show all calculations, direct answers are not acceptable. Your answers should be approximated up to two decimal places. Attempt all parts of a question together.

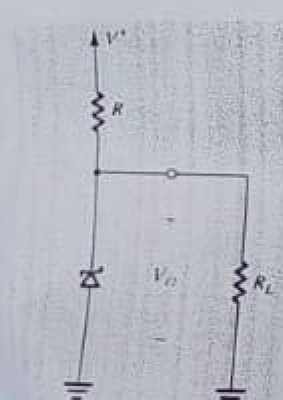
Question: 1 (CLO # 2) **10 +10= 20 Marks**

1) Draw the circuit inside the black box which has the transfer characteristics as given below.





2) A Zener shunt regulator employs a 9.1V Zener diode for which $V_z = 9.1V$ at $I_z = 9mA$, with $r_z = 30\Omega$ and $I_{ZK} = 0.3mA$. The available supply voltage has a nominal value of 15V. For this diode



i. What is the value of V_{Z0}?

ii. For a nominal load resistance R_L of $1k\Omega$ and a nominal Zener current of 10mA, what current must flow in the supply resistor R?

iii. For the nominal value of supply voltage, select a value of resistor R to provide the minimum current?

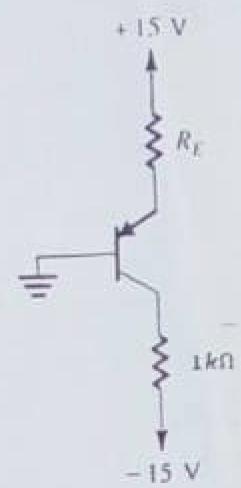
iv. Find the line regulation?

15 Marks

Question: 2 (CLO # 4)

For the following circuit having a BJT with $\beta = 100$, select a value of R_E such that

a) BJT saturates with a forced β of 10.
b) BJT is at the edge of saturation.



15 Marks

Question: 3 (CLO # 5)

A common base amplifier is required to amplify a signal delivered by a co-axial cable having a source resistance of 50Ω. What bias current I_C should be utilized to obtain R_{in} that is matched (equal) to the cable resistance? To obtain an overall voltage gain of G_V of 40 V/V, what should the total resistance in the collector be? Use $\beta = 200$.